ABU DHABI CYCLING NETWORK

BASIS OF DESIGN

JULY 2022

ALDAR

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1 INTRODUCTION

1 INTRODUCTION

OVERVIEW

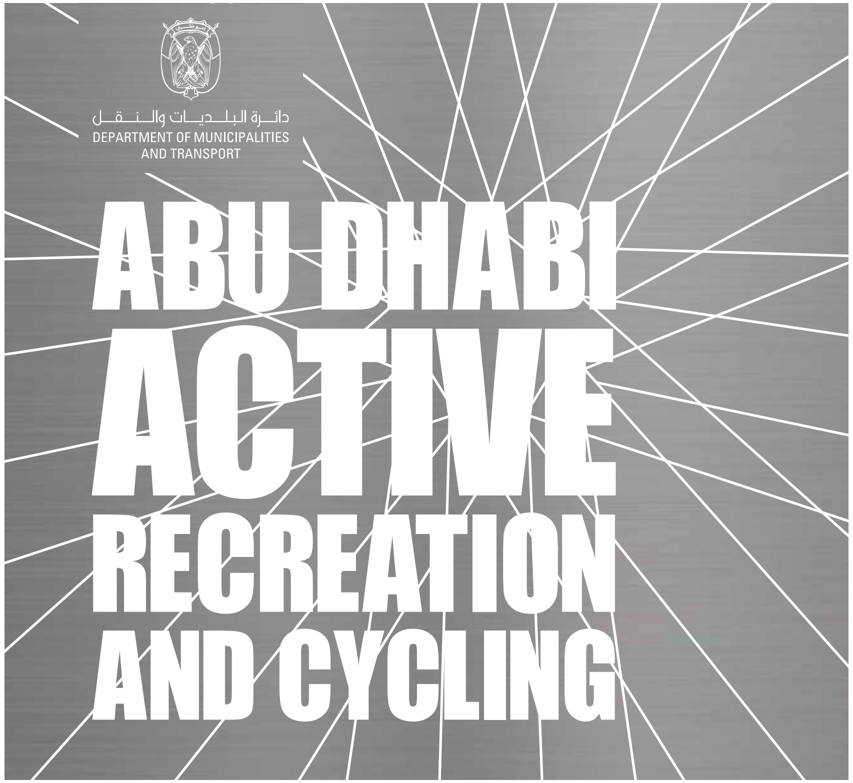
This document, prepared for Aldar by Design Consultants, represents the Basis of Design Report for the proposed Abu Dhabi Cycling Network (ADCN). The Basis of Design sets out the key planning principles used in the development of the ADCN.

The original scheme and proposals for the ADCN follow the vision set out in the Abu Dhabi Active Recreation and Cycling Strategy, produced by the Department of Municipalities and Transport (DMT) in November 2021.

In 2021, the Emirate was awarded the prestigious Union Cycliste Internationale (UCI) 'Bike City' label, the first in Asia, supporting the long-term vision to become a leading global cycling hub. This label recognises Abu Dhabi's achievements to promote cycling and further expand on the Emirate's cycling infrastructure and programmes. It also supports future hosting of key international UCI events, including the 2022 and 2024 UCI Urban Cycling World Championships, as well as the 2028 UCI Gran Fondo World Championships.

These events will promote global cycling competition and as well as offering a chance to showcase Abu Dhabi's investments and initiatives in recreational, utility and sports cycling to the World.

Abu Dhabi joins other established 'Bike Cities' including Bergen, Copenhagen, Glasgow, Paris, Vancouver and Yorkshire among others and this is expected to drive further investment and a step-change in cycling provision in the next five years.



THE VISION FOR CYCLING IN ABU DHABI, SET OUT IN THE ACTIVE RECREATION AND CYCLING STRATEGY, IS TO:

"ESTABLISH ABU DHABI AS A GLOBAL LEADER FOR CYCLING, IN SPORT, LEISURE AND MOBILITY, HELPING THE POPULATION ENGAGE IN A HEALTHIER LIFESTYLE."

This vision will be delivered through 7 overarching principles as set out in Figure 1. These principles in turn are met through the planning and delivery of cycling infrastructure and facilities for recreation, utility, and sports cycling, as well as well other complementary measures, including public education, training, and behavioural change.

The vision is therefore much broader than the cycling infrastructure and facilities which will be planned, designed, and delivered through the ADCN aiming for Abu Dhabi to become a leading global cycling hub, with world-class tracks, international events and cycling venues.

Through a new platform and brand - Bike Abu Dhabi – the aim is to promote cycling growth as a health and leisure activity, as a green and sustainable means of transport, as a competitive sport for amateurs and professionals, as a culture and as an enabler of a happier, more inclusive, and liveable city.

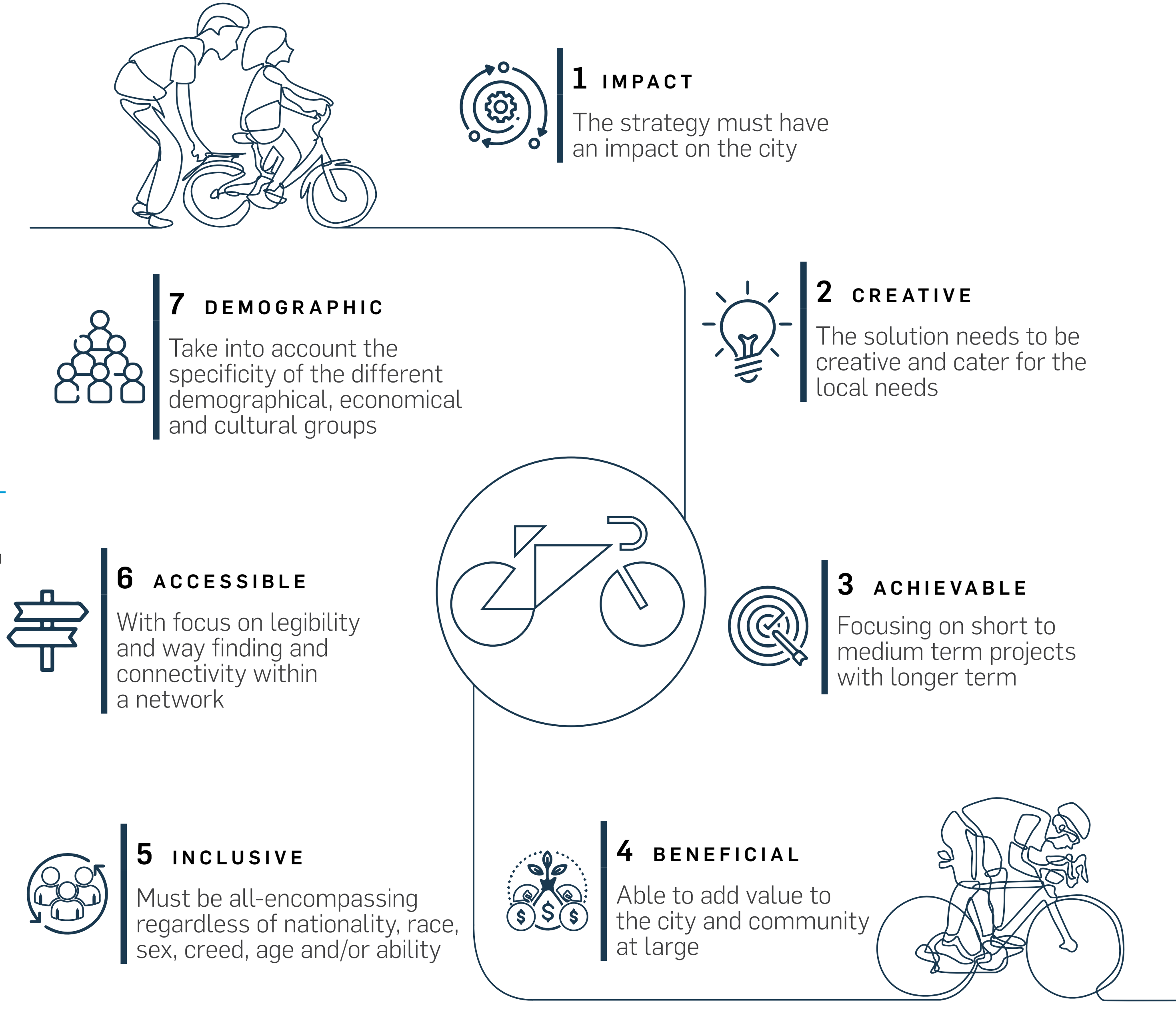


Figure 1: Active Recreation and Cycling Principles

USERS

Before planning and designing cycling infrastructure and facilities, user group categories must be identified, defined, and understood, to ensure that these are met adequately through the ADCN design process.

Considering differing design requirements, four overarching categories of users are envisioned to be using the ADCN as set out in Figure 2, aligned to the vision.

The user groups identified have several common and differing needs as shown in Figure 3.

Recreational Cyclists

Local residents and tourists of all age groups, including families with children and older people, with various cycling abilities and skills and who want to use bicycle routes for outdoor leisure in order to have fun, exercise, socialise with others and have a safe and wondrous introduction to cycling.

Explorers

Local residents and tourists with low or moderate cycling skills and experience and who want to use bicycle routes to enjoy scenic views and travel to attractive destinations in order to explore the city, facilities and its surroundings.

Utility Cyclists

Locals with moderate or advanced cycling skills and who want to use bicycle routes to travel to work, educational institutions, shopping and conduct other personal business in order to combine physical exercise with savings on transportation costs, often under time constraints.

Sports Cyclists

Local residents and tourists – serious and advanced cyclists - who want to use bicycle routes for sport and exercise in order to improve their physical fitness or cycling performance, including competition.

Figure 2: User Groups of the Abu Dhabi Cycling Network

Recreational Cyclists 

Explorers 

Utility Cyclists 

Sports Cyclists 

- **Safe Routes**: Bicycle routes are planned and designed to reduce the frequency and severity of crashes, limit conflicts between users and convey a perception of user safety, including from general traffic and pedestrians.
- **Comfortable Routes:** Bicycle routes ensure that cyclist experience minimal stops, nuisance or inconvenience and have confidence in the route ahead.
- **Attractive Routes:** Bicycle routes are easily accessible (e,g., car parking, bicycle rental) and designed, furnished (e,g., bicycle parking, access to fresh water, restrooms facilities) and illuminated with personal safety in mind to make cycling socially safe and attractive.

Recreational Cyclists

- Accessible for all cyclist
- Segregated from the motorised traffic
- Short-term bicycle parking
- Capacity to accomodate slow and inexperienced cyclists

Explorers

- Frequent resting spots
- Wayfinding, local information and route signage
- Bicycle rental spots
- Access to drinking water

Utility Cyclists

- Direct route
- Safe long-term parking
- Minimised numbers of stops
- Connection to and integration with public transport

Sports Cyclists

- Fast direct routes
- Minimised potential conflicts with pedestrians
- Access to drinking water
- Smooth, well-maintained pavement

Figure 3: Common and Contrasting Requirements for User Groups

ROUTES WITHIN THE ABU DHABI CYCLING NETWORK

The routes to be planned and designed within the ADCN are described below and illustrated in Figure 4.

Abu Dhabi Island At-Grade Network - 193 Km of at-grade cycle track aligned along the main street corridors and secondary routes of Abu Dhabi Island. (see pages 14-15)

Abu Dhabi Island High-Speed Cycle Track (HSCT) - 47 Km elevated loop around Abu Dhabi Island and connecting Al Reem and Al Mariyah Islands. (see pages 16-17)

Abu Dhabi Mainland (Connection of AD Island to Yas Island and Saadiyat Island) - 62 Km at-grade route connecting to Abu Dhabi Island at the Sheikh Khalifa Bridge, continues through Saadiyat and Yas Islands and onward toward Al Raha. The route loops back to Abu Dhabi Island via Musaffah and Maqta'a Bridges. (see pages 18-19)

Abu Dhabi Mainland to Abu Dhabi Boundary (Al Qudra) - 87 Km at-grade route from Yas Island towards KIZAD, connecting with the Al Qudra Cycle Track. (see pages 20-21)

Abu Dhabi Mainland to South of Baniyas (Al Wathba) - 42 Km at-grade route adjacent to the Abu Dhabi-Al Ain Road toward the Al Wathba area. (see pages 22-23)

In the following pages routes are explored with a reference to the route colours assigned here.

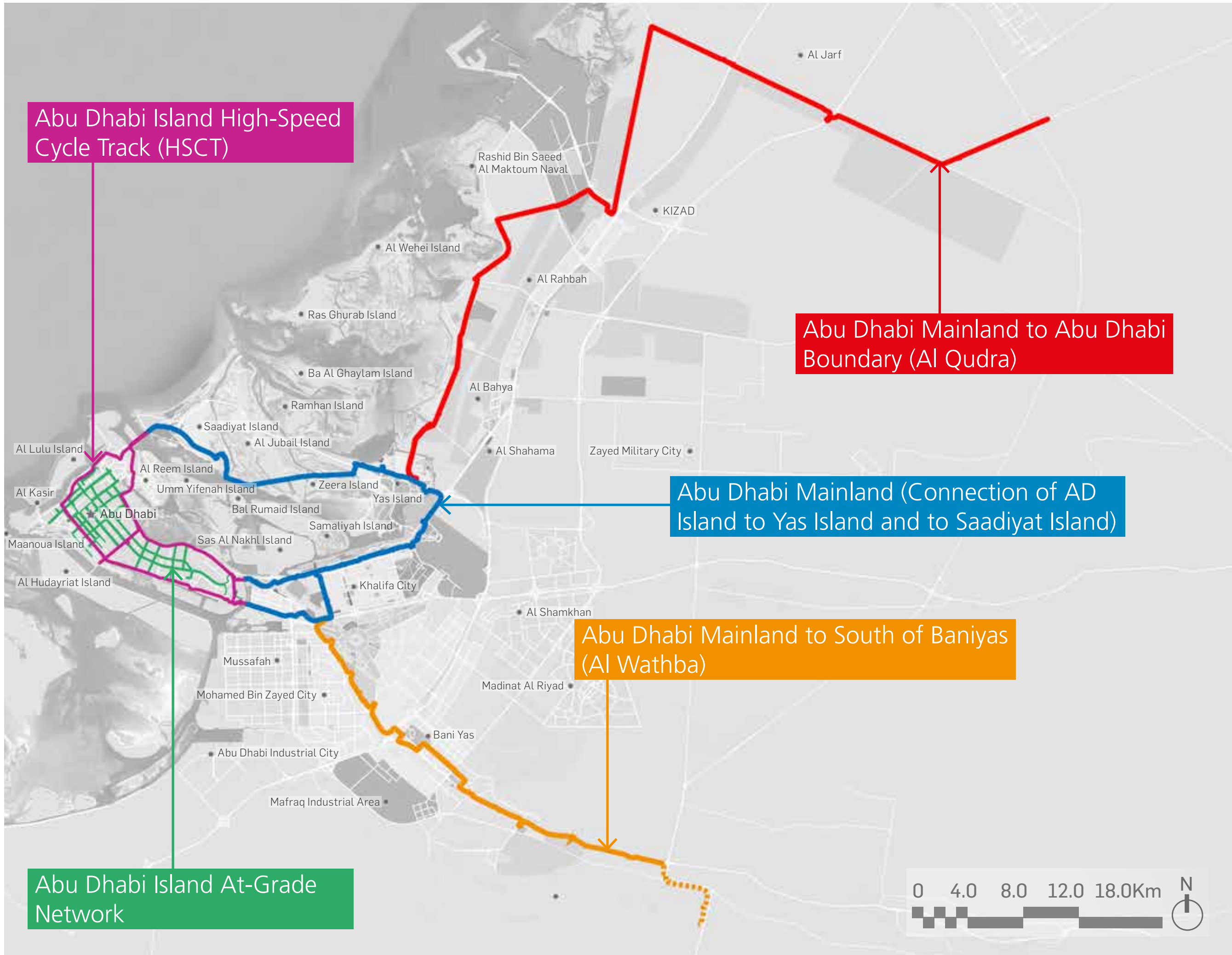


Figure 4: Route Alignments within the Abu Dhabi Cycling Network

FUNCTION

The user group definitions and their corresponding requirements have informed the development of route typologies i.e., the classification of sections of each route by the combination of user groups most likely to be utilising each respective section.

Three route typologies have been established and Figure 5 illustrates the route typologies assigned to the respective sections of each route.

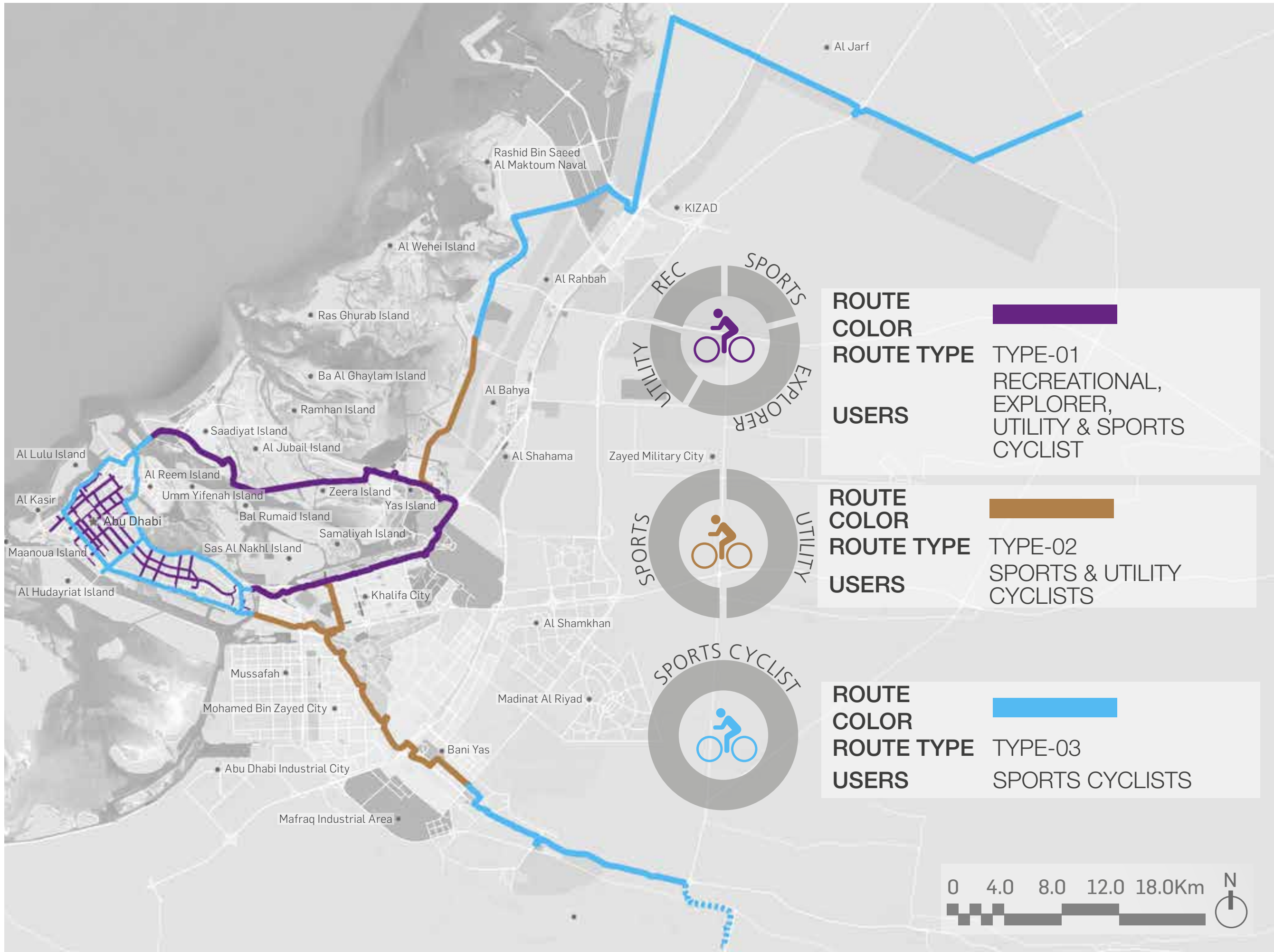


Figure 5: Route Typologies within the Abu Dhabi Cycling Network

2 DESIGNING THE VISION

2 DESIGNING THE VISION

APPLICABLE STANDARDS

The planning and design of the key cycling routes to Abu Dhabi is based on specific engineering and architecture standards to ensure application of state-of-the-art cycling infrastructure planning practices, and additional international standards and guidelines are also referred to.

The standards, codes, and guidelines to be considered during the planning and design of the key routes within the ADCN are detailed in Appendix B. Additionally, the references for the route-specific design parameters are listed in Appendix C.

OVERARCHING DESIGN PRINCIPLES

There is a need to define overarching principles that set the basis for the planning and design of the ADCN.

The design principles promote cycling, which is accessible to all, irrespective of age, sex, culture, or income, and within the geographical, climatic, and cultural context of Abu Dhabi. They cover the core and supporting aspects of cycling infrastructure, which in turn are to be translated into tangible assets that embody these directions on the ground.

Figure 6 shows the 15 overarching design principles for the ADCN as developed iteratively between DMT, Aldar and the Design Consultants.

EMERGENCY ACCESS

Emergency access is a key consideration during development and operation of the ADCN and needs to consider ambulance, civil defence and police vehicles and personnel.

The at-grade cycle routes on Abu Dhabi Island and on the Mainland are aligned parallel to the road edge for most of their alignment with a maximum distance of 25m from the road edge, allowing for ease of emergency vehicle access.

In the mangrove section of the Abu Dhabi to Yas and Saadiyat Island route, access to the cycle track is obtained from the nearest junction of the cycle route with the roadway. For the High-Speed Cycle Track on Abu Dhabi Island, emergency vehicles may reach the elevated track via access ramp. Alternatively, a provisional vehicle could be kept on standby mode on the track itself in a dedicated location.

ROUTE-SPECIFIC DESIGN PRINCIPLES

Tables 1-5 and Figures 7-16 define the design parameters to be adopted during the planning and design of each route within the ADCN. Whilst following applicable standards (see Appendices B and C), certain parameters such as width, vertical alignment, spacing and design of rest areas, are specific to individual routes, while other parameters such as lighting, wayfinding and signage principles are generally common.

In some cases, to incorporate the cycle route within the available space, right of way or other local circumstances, it may be necessary to deviate from the applicable standards or desired design approach.

CONNECTED AND CONTINUOUS

Cycling routes and facilities connect key origins and destinations -anywhere to everywhere- through continuous rights of way and safe intersections, without barriers, detours, gaps, and missing links which might inconvenience and slow cyclists down or require them to dismount unnecessarily.

CONVENIENT AND CONVIVIAL

Cyclists find it easy, comfortable, smooth, pleasurable, convenient, aesthetically pleasing, well maintained and enjoyable to locate, reach, use and experience the cycle routes and facilities and the opportunities which are enabled.

DIRECT AND AVOIDING UNNECESSARY DIVERSIONS

Cycling routes and facilities directly connect key destinations avoiding unnecessary diversions and deviations, providing grade separated crossings thus minimising delays and creating an attractive alternative to car trips. Direct routes are logical and intuitively understandable by all road users.

ACCESSIBLE AND USER FOCUSED

Cycling routes and facilities are conveniently located, easily found, and accessed and used by a wide range of user types of varying ages, fitness levels and abilities with infrastructure designed around understanding, and seeking to meet, their needs and expectations, as well as wider community needs.

SAFE AND SECURE

Cyclists are, and feel, safe when using the cycle routes and facilities, especially from road traffic, but also managed in terms of their impacts on, and potential conflicts with, pedestrians and other users of the public realm, as well as being protected from the threat of crime, such as cycle theft or vandalism.

INCLUSIVE AND INTEGRATED

Cycling routes and facilities are open to a range of user types, including people of determination, and are integrated with adjacent land uses, developments, mobility hubs and other transport modes, including access by private cars, and shared and public transport.

NAVIGABLE AND LEGIBLE

Cycling routes and facilities are coherent, visible, well signposted and marked, provide easy to interpret directions and guidelines for users to follow, and enhance the cycling experience with essential, supportive, and context-specific information and wayfinding.

SUSTAINABLE AND CARBON NEUTRAL

Cycling routes and facilities are designed to conserve, enhance, or protect the natural and built environment, promote green open spaces, minimise impacts on natural ecosystems, resources, and landscapes, and seek to reduce embedded or operational energy intensity and carbon emissions.

Figure 6: Overarching Design Principles for the ADCN (All Routes)

CLIMATE APPROPRIATE AND RESILIENT

Cycling routes and facilities are designed to be accessible and useable throughout as much of the year as possible, including adaptability to weather conditions, and providing effective passive or active cooling at key nodes and facilities to allow cycling during the warm summer months.

COMPLIANT

Cycling routes and facilities are planned based on all applicable and relevant planning and design standards, guidelines and regulations issued by the competent authorities in the Emirate and the UAE.

ENABLING CULTURE AND MEMORABLE EXPERIENCES

Cycling routes and facilities reflect and embody the cultural, entertainment, sports and recreational assets, opportunities, and experiences to which they provide access and enable.

UNIQUE AND REFLECTING BRAND ABU DHABI

Cycling routes and facilities are designed to be iconic, distinctive and provide a visible symbol and representation of the Emirate and all it has to offer residents, tourists, and visitors.

INNOVATIVE AND SMART

Cycling routes and facilities may be designed to test, demonstrate, or deploy new technologies, concepts and innovations for the greater promotion, take-up, and benefits of cycling and other forms of active travel and micromobility, including projecting the Emirate as a centre for smart mobility infrastructure and services.

ACHIEVABLE AND DESIRABLE

Proposals for cycling routes and facilities must be practical, able to be easily constructed, operated, and maintained, and demonstrably achieve their strategic objectives to increasing cycle use and mode share within the Emirate.

TARGET COST AND VALUE FOR MONEY

Cycling routes and facilities must be designed within a cost envelope, must ensure that the benefits are monitored and shown to be justified by the resources and time expended on their implementation.

Figure 6: Overarching Design Principles for the ADCN (All Routes),...continued

Table 1: Design Parameters for Abu Dhabi Island At-Grade Cycle Network

DESIGN PARAMETERS[1]	
User Groups	**Recreational, Explorers, Utility and Sports Cyclists**
Route Length	193 Km
Track Width (Desirable)	3m
Track Width (Minimum)	2m **[1]**
Design Speed	20 Kph
Stopping Sight Distance	24m **[2]**
Minimum Radius	8m **[3]**
Vertical Clearances	Above track finished level - 2.3m - 2.5m. **[4]**
Interface Priority between Other Users	• At interface between cyclist and pedestrian - priority for pedestrian. • At road crossing - priority for pedestrian and cyclist. • Cyclists and vehicular shared lanes are not permitted.
Surface Type	• In line with Abu Dhabi standards. • Tiles at location were requested by local authorities to protect the underground utilities. • Paving of shared areas - ADM paving strategy **[5]**
Gradients	2% (Max) **[6]**
Horizontal Clearance	• 0.5m (min) clearance from edge of track to adjacent trees, poles, signposts, etc. **[7]** • 0.5m (min) buffer between cycle track and pedestrian walkways. **[7]** • Light poles, signposts, cabinets, trees, parking metres – not allowed within the cycle track path.
Requirements, and Frequency of Rest Stops	• Cycle track connects to at-grade facilities along the way, such as bus stops, public shared areas, parks, mosques. **[8]** • 6 cycle racks, able to accommodate up to 12 bicycles, to be provided at bus stops, parks and mosques.
Lighting Levels	• 3 – 5 Lux (Average horizontal illumination) **[9]** • 1.5 – 2 Lux (Average vertical illumination) **[9]**
Utility, Signage & Tree Adjustment or Relocation	• Signposts, light poles, parking pay and display, cabinets, pillar fire hydrants to be relocated. • Levels of existing utility chambers within cycle track or pedestrian/cyclist shared area to be adjusted (raised or lowered) to match finished top level of cycle tracks/shared areas. • Hedges, groundcover, grass, trees, palms, etc. and associated existing irrigation network within cycle tracks or pedestrian/cyclist shared areas to be removed/relocated. • Cycle track signage at major road crossings and every 500m interval. **[10]**
Distance of Warnings Before Crossings	Warning signs - at start of shared space arrangement **[11; 12; 13]**
Road Crossing	• At grade crossing - 7m (min 4m) with dropped curbs **[14]** • At bus stops - 10m (min) shared area **[15]** • Shared area at major road junction - 10m (min) from edge of crossing **[16]** • Shared area at minor road junctions - 5m (min) from edge of crossing **[16]** • At road crossings - tactile paving and bollards **[17]**
Other Specific Design Details	The cycle track to be diverted away from existing poles, deep sewer manholes, fire hydrants, high-pressure water mains chambers, 132Kv; 220Kv & 400Kv power transmission cables, etc.

[1]*See Appendix C for the List of References*

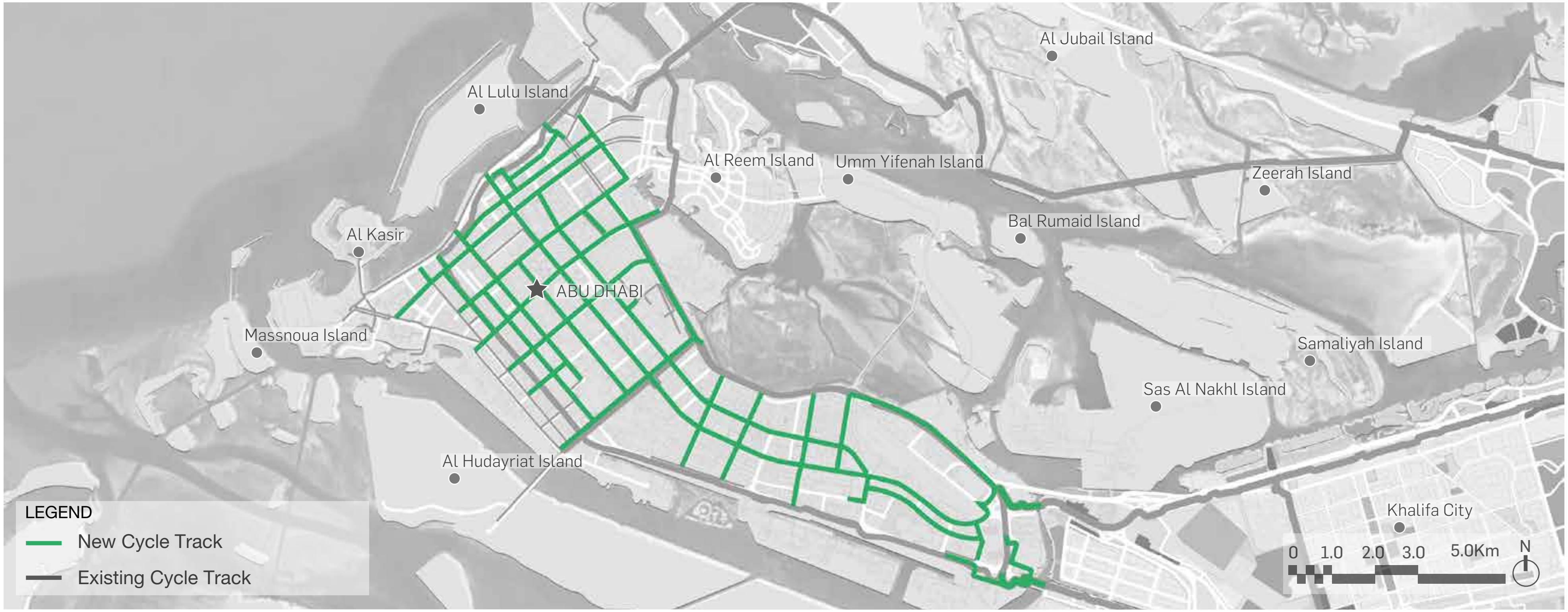


Figure 7: Route Alignment of Abu Dhabi Island At-Grade Cycle Network

Figure 8: Illustrative Sketches of Abu Dhabi Island At-Grade Cycle Network

Table 2: Design Parameters for Abu Dhabi Island High-Speed Cycle Track

DESIGN PARAMETERS	
User Groups	**Sports Cyclists.** Potential intermittent use by Utility cyclists during peak hours.
Route Length	47 Km
Track Width (Desirable)	7m
Track Width (Minimum)	7m
Design Speed	40 Kph
Stopping Sight Distance	67m **[18]**
Minimum Radius	40m **[19]**
Vertical Clearances	Elevated track. Vertical clearance: • Crossing above roads - 6.5m clearance below the track. • When not crossing roads - 3.5m clearance below the track. **[20]** • Water bodies - navigational clearance (6m - 11m) above Highest Astronomical Tides level **[21]**
Interface Priority between Other Users	• Commuters travelling at 10 - 20 Kph and during business commuting hours to be considered as an intermittent user group. • Emergency vehicle for rescue to use ramps when needed. • Interference managed through ITS – gate control systems, registered mobile applications, digital signs, etc.
Surface Type	• In line with Abu Dhabi standards.
Gradients	• 2.5% **[22]** • Ramps: max 5 % **[23]** with landing of 25m length at every 3m vertical ascent. **[24]**
Horizontal Clearance	• 10m **[25]** • 2m high privacy screen where minimum horizontal clearance is not achievable. **[26]**
Requirements, and Frequency of Rest Stops	• Rest stops not considered as a requirement. • Accesses/Exits ramps to the sides of the track at an average interval of 6 Km.
Lighting Levels	• 8 – 10 Lux (Average horizontal illumination) **[9]** • 3 – 5 Lux (Average vertical illumination) **[9]** • Lighting below the cycle track at identified locations to be considered.
Utility, Signage & Tree Adjustment or Relocation	• Integrated Signage and Wayfinding strategy • A dedicated storm water drainage system. • Dedicated CCTV (MCC) system to monitor all access and usage of the track and access ramps. • A dedicated ITS capable of managing speed, direction of flow on the cycle track. • Tree removal and/or relocation to be kept to the minimum. • Impacts on existing traffic assets including traffic lights, street lighting, signage, CCTV (MCC), and gantry systems to be minimized.
Distance of Warnings Before Crossings	Not applicable
Road Crossing	Not applicable
Other Specific Design Details	• 1.5m (min) high railing **[27]** • The cycle track to include a dedicated mobile application to manage the usage of the track. • Modular / prefabrication construction • Estidama 2 pearls and LEED silver are targeted sustainability targets

[1] *See Appendix C for the List of References*

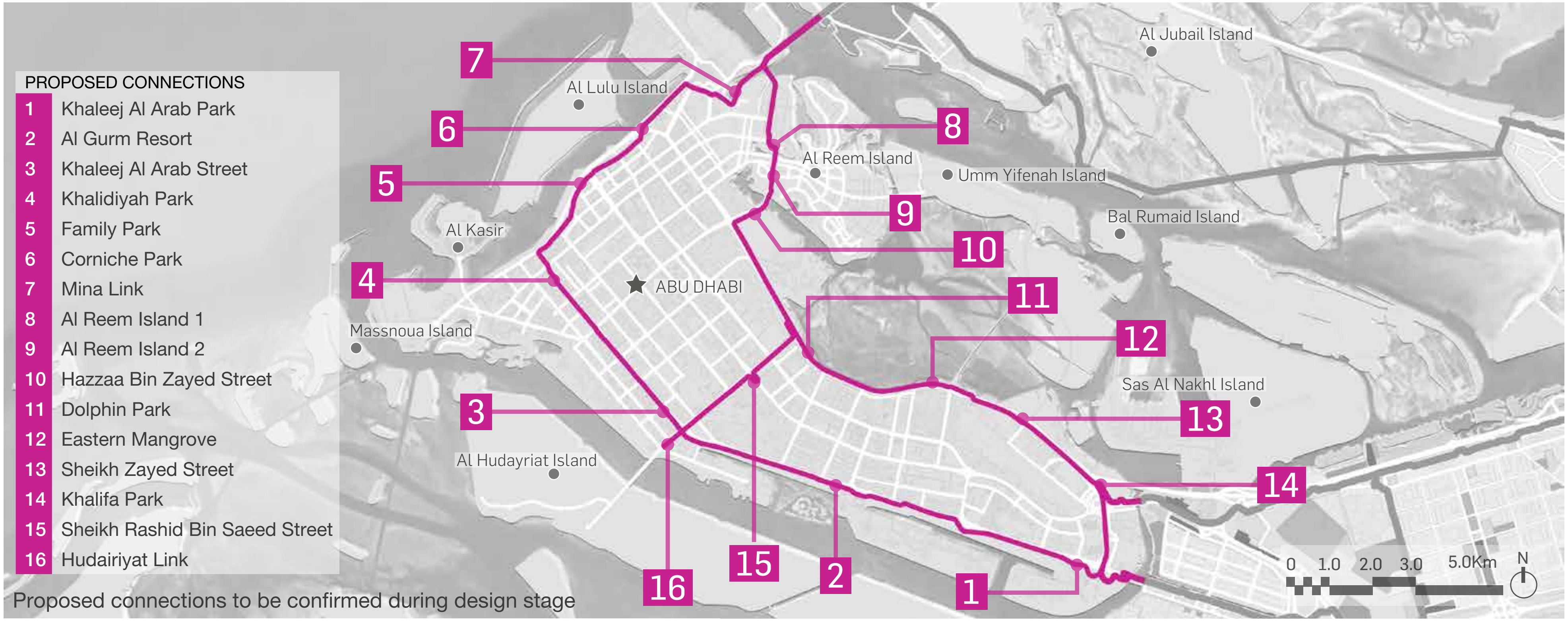


Figure 9: Route Alignment of Abu Dhabi Island High-Speed Cycle Track

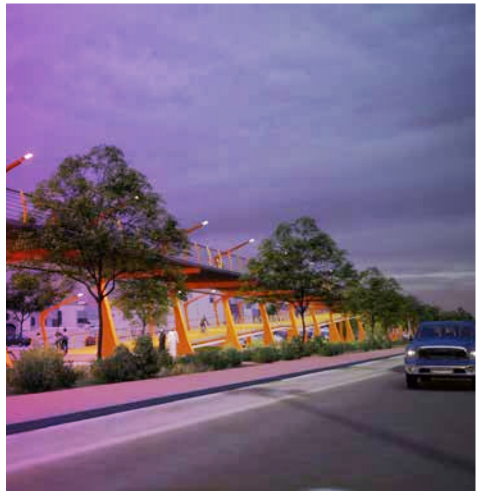


Figure 10: Illustrative Sketches of Abu Dhabi Island High-Speed Cycle Track (DMT-AD Cycling Strategy 01/06/21)

Table 3: Design Parameters for Abu Dhabi Mainland (connection of AD Island to Yas Island and to Saadiyat Island) Cycle Route

DESIGN PARAMETERS[1]	
User Groups	**Recreation, Utility and Sport Cyclists**
Route Length	62 Km
Track Width (Desirable)	7m
Track Width (Minimum)	3m (17 % of the route is < 7m wide) **[28]**
Design Speed	20 Kph – Urban & Residential Areas, 40 Kph – Rural
Stopping Sight Distance	20 Kph – 24m, 40 Kph – 67m **[29]**
Minimum Radius	20 Kph – 9m, 40 Kph – 35m **[30]**
Vertical Clearances	• Crossing above roads – 6.5m **[31]** • Underpasses & tunnel – 3.5m **[31]** • Water bodies - navigational clearance (6m - 11m) above Highest Astronomical Tides level **[21]**
Interface Priority between Other Users	• At interface between cyclist and pedestrian - priority for pedestrian. • At road crossing - priority for pedestrian and cyclist. • Cyclists and vehicular shared lanes are not permitted.
Surface Type	• In line with Abu Dhabi standards. **[32]** • Use of tiles at specific location were requested by local service provider authority to protect the underground utilities. • Paving of shared areas - ADM paving strategy **[5]**
Gradients	5% (Max vertical grade for ramps) **[33]**
Horizontal Clearance	• 1.5m (min) buffer from the road edge to the cycling track with provision space for landscape. **[28]** • 2m (min) buffer with provision for tree corridor. **[28]** • 60cm clearance for signs. **[34]**
Requirements, and Frequency of Rest Stops	• 10 Km (max) between rest stops and hydration facilities. • Bigger nodes at the start/end points of the route.
Lighting Levels	• 3 – 5 Lux (Average horizontal illumination) **[35]**
Utility, Signage & Tree Adjustment or Relocation	• Existing infrastructure utilities to be protected with concrete protection slab. • Levels of existing chambers within cycle tracks or pedestrian/cyclist shared areas to be adjusted (raised or lowered) to match finished top level of the cycle tracks or shared areas. • Signposts, light poles, parking pay and display, cabinets, pillar fire hydrants, etc. within cycle tracks path to be relocated • Existing Ghaf tree to be retained as much as possible • Trees affect by the routes to be relocated and replanted within the project
Distance of Warnings Before Crossings	Warning signs - at start of shared space. **[11; 12; 13]**
Road Crossing	• At crossing of major road junctions, shared area - minimum 10m from edge of crossing **[16]** • At crossing of minor road junctions, shared area - minimum from 5m edge of crossing **[16]** • At grade crossing width - 7m (minimum 4m) with dropped curbs **[14]** • At bus stops, shared areas - 10m minimum width **[15]** • Tactile Paving and bollards be provided at road crossings with shared area. **[17]** • At crossing in rural area cycle only crossings are proposed with 'SLOW' text marking & red colour pavement 5m before road crossing. **[13]**
Other Specific Design Details	• Track above existing Transco corridor within the mangrove area to be asphalt (except at Location of the manholes cover for underground utilities to be tiles) • Any structure footing and bridge ramp to be located away from the existing Transco corridor

[1] *See Appendix C for the List of References*

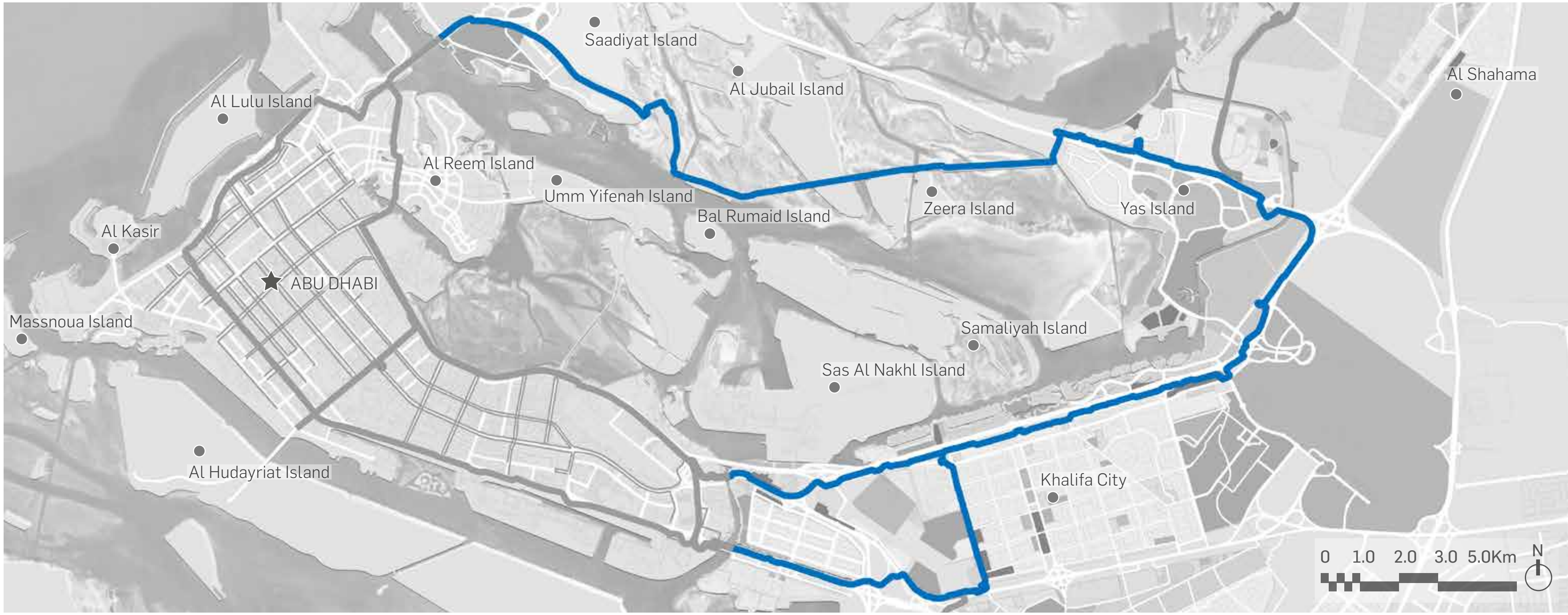


Figure 11: Route Alignment of Abu Dhabi Mainland (connection of AD Island to Yas Island and to Saadiyat Island)

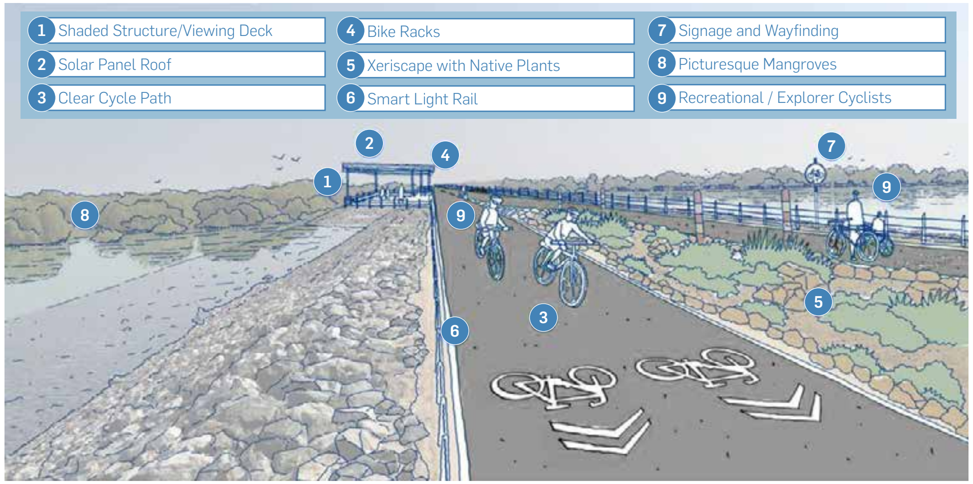


Figure 12: Illustrative Sketches of Abu Dhabi Mainland (connection of AD Island to Yas Island and to Saadiyat Island) Cycle Route

Table 4: Design Parameters for Abu Dhabi Mainland to Abu Dhabi Boundary (Al Qudra) Cycle Route

DESIGN PARAMETERS	
User Groups	**Sport and Utility Cyclists**
Route Length	87 Km
Track Width (Desirable)	7m
Track Width (Minimum)	3m (35 % of the route is < 7m wide) **[28]**
Design Speed	20 Kph – Urban & Residential Areas, 40 Kph – Rural
Stopping Sight Distance	20 Kph – 24m, 40 Kph – 67m **[29]**
Minimum Radius	20 Kph – 9m, 40 Kph – 35m **[30]**
Vertical Clearances	• Crossing above roads – 6.5m **[31]** • Underpasses & tunnel – 3.5m **[31]** • Water bodies - navigational clearance (6m - 11m) above Highest Astronomical Tides level **[21]**
Interface Priority between Other Users	• At interface between cyclist and pedestrian - priority for pedestrian. • At road crossing - priority for pedestrian and cyclist.
Surface Type	• In line with Abu Dhabi standards. **[32]** • Use of tiles at specific location were requested by local service provider authority to protect the underground utilities. • Paving of shared areas - ADM paving strategy **[5]**
Gradients	5% (Max vertical grade for ramps) **[33]**
Horizontal Clearance	• 1.5m (min) buffer from the road edge to the cycling track with provision space for landscape. **[28]** • Ghanadhah street is an exception - buffer from road is 1m. • 2m (min) buffer with provision for tree corridor. **[28]** • 60cm clearance for signs. **[34]**
Requirements, and Frequency of Rest Stops	• 10 Km (max) between rest stops and hydration facilities. • Bigger nodes at the start/end points of the route
Lighting Levels	3 – 5 Lux (Average horizontal illumination) **[35]**
Utility, Signage & Tree Adjustment or Relocation	• Existing infrastructure utilities to be protected with concrete protection slab. • Levels of existing chambers within cycle tracks or pedestrian/cyclist shared areas to be adjusted (raised or lowered) to match finished top level of the cycle tracks or shared areas. • Signposts, light poles, parking pay and display, cabinets, pillar fire hydrants, etc. within cycle tracks path to be relocated • Existing Ghaf tree to be retained as much as possible • Trees affect by the routes to be relocated and replanted within the project
Distance of Warnings Before Crossings	Warning signs - at start of shared space **[11; 12; 13]**
Road Crossing	• At grade crossing - 7m (min 4m) with dropped curbs **[14]** • At bus stops - 10m (min) shared area **[15]** • Shared area at major road junction - 10m (min) from edge of crossing **[16]** • Shared area at minor road junctions - 5m (min) from edge of crossing **[16]** • At road crossings - tactile paving and bollards **[17]** • At crossing in rural area cycle only crossings are proposed with 'SLOW' text marking & red colour pavement 5m before road crossing. **[13]**

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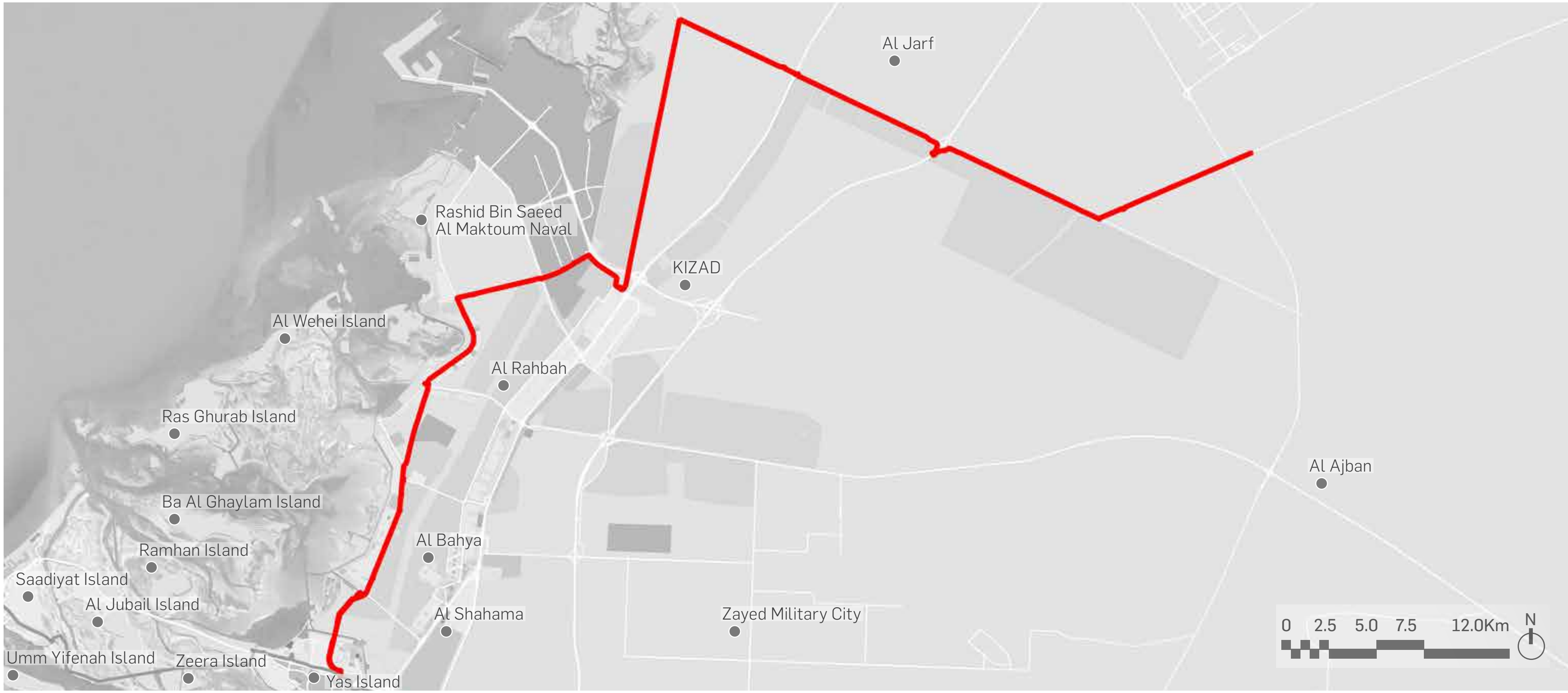


Figure 13: Route Alignment of Abu Dhabi Mainland to Abu Dhabi Boundary (Al Qudra)

Figure 14: Illustrative Sketches of Abu Dhabi Mainland to Abu Dhabi Boundary (Al Qudra) Cycle Route

Table 5: Design Parameters for Abu Dhabi Mainland to South of Baniyas (Al Wathba) Cycle Route

DESIGN PARAMETERS	
User Groups	**Sport and Utility Cyclists**
Route Length	42 Km
Track Width (Desirable)	7m
Track Width (Minimum)	3m (40 % of the route is < 7m wide) **[28]**
Design Speed	20 Kph – Urban & Residential Areas, 40 Kph – Rural
Stopping Sight Distance	20 Kph – 24m, 40 Kph – 67m **[29]**
Minimum Radius	20 Kph – 9m, 40 Kph – 35m **[30]**
Vertical Clearances	• Crossing above roads – 6.5m **[31]** • Underpasses & tunnel – 3.5m **[31]**
Interface Priority between Other Users	• At interface between cyclist and pedestrian - priority for pedestrian. • At road crossing - priority for pedestrian and cyclist.
Surface Type	• In line with Abu Dhabi standards. **[32]** • Use of tiles at specific location were requested by local service provider authority to protect the underground utilities. • Paving of shared areas - ADM paving strategy **[5]**
Gradients	5% (Max vertical grade for ramps) **[33]**
Horizontal Clearance	• 1.5m (min) buffer from the road edge to the cycling track with provision space for landscape. **[28]** • 2m (min) buffer with provision for tree corridor. **[28]** • 60cm clearance for signs. **[34]**
Requirements, and Frequency of Rest Stops	• 10 Km (max) between rest stops and hydration facilities. • Bigger nodes at the start/end points of the route
Lighting Levels	3 – 5 Lux (Average horizontal illumination) **[35]**
Utility, Signage & Tree Adjustment or Relocation	• Existing infrastructure utilities to be protected with concrete protection slab. • Levels of existing chambers within cycle tracks or pedestrian/cyclist shared areas to be adjusted (raised or lowered) to match finished top level of the cycle tracks or shared areas. • Signposts, light poles, parking pay and display, cabinets, pillar fire hydrants, etc. within cycle tracks path to be relocated • Existing Ghaf tree to be retained as much as possible • Trees affect by the routes to be relocated and replanted within the project
Distance of Warnings Before Crossings	Warning signs - at start of shared space per **[11; 12; 13]**
Road Crossing	• At grade crossing - 7m (min 4m) with dropped curbs **[14]** • At bus stops - 10m (min) shared area **[15]** • Shared area at major road junction - 10m (min) from edge of crossing **[16]** • Shared area at minor road junctions - 5m (min) from edge of crossing **[16]** • At road crossings - tactile paving and bollards **[17]** • At crossing in rural area cycle only crossings are proposed with 'SLOW' text marking & red colour pavement 5m before road crossing. **[13]**

[1] *See Appendix C for the List of References*

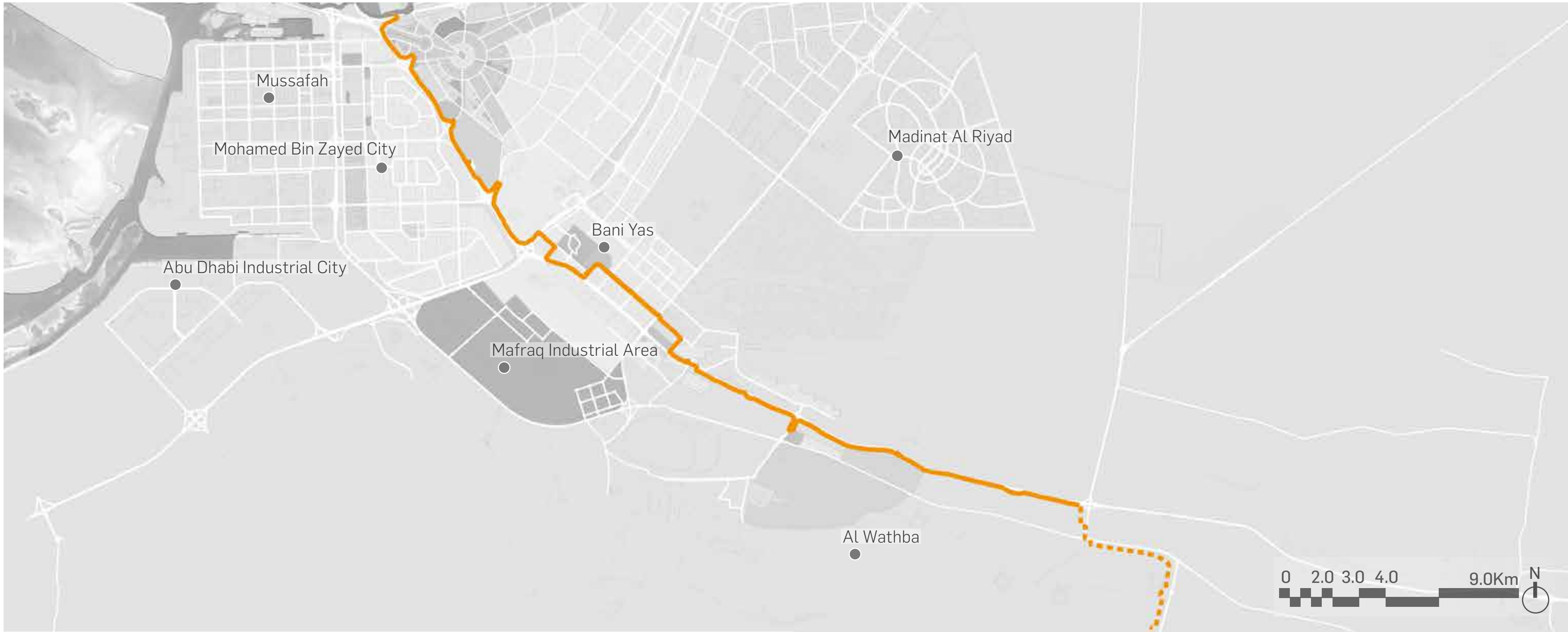
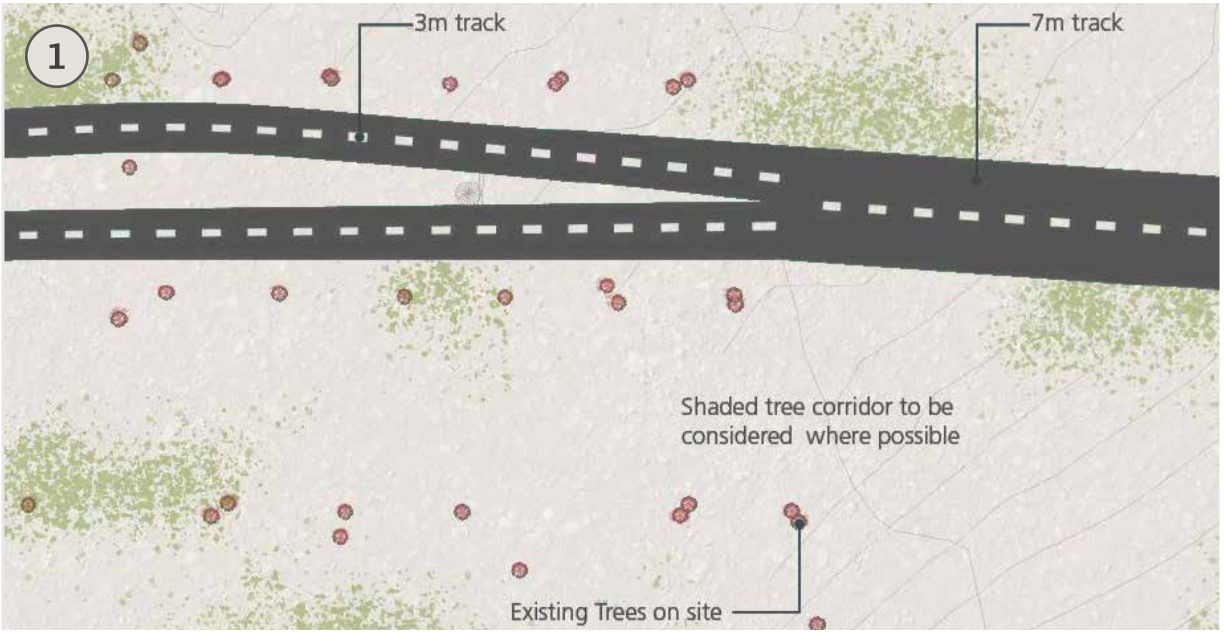


Figure 15: Route Alignment of Abu Dhabi Mainland to South of Baniyas (Al Wathba)

Figure 16: Illustrative Sketches of Abu Dhabi Mainland to South of Baniyas (Al Wathba) Cycle Route

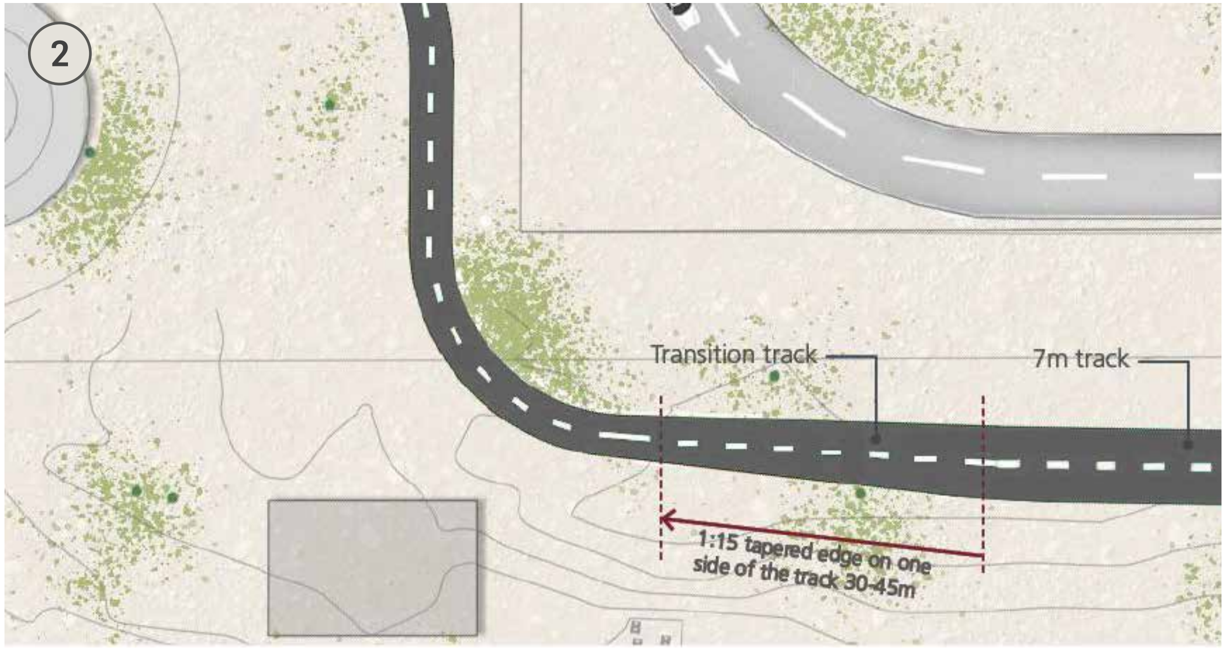
ALIGNMENT STRATEGIES

The enlargement plans below show typical strategies to adjust the cycling track alignment to fit existing conditions, including existing roads, crossings, junctions, utility corridors, trees and other circumstances. There are five typical strategies which are being deployed to the design of the ADCN.



1. CYCLE TRACK SPLIT

The cycling track may be split from the desired bi-directional 7 metre width to two 3m wide one-way tracks at certain locations. This has been done not only to avoid conflict with the existing trees and other obstacles on site where these cannot be moved, but also to improve cycling experience within natural or artificial landscaped areas and offer pleasant cycling journey.



2. CYCLING TRACK WIDTH TRANSITION

Where there are constraints due to a narrow right of way or conflicts with existing utility corridors, the width of the cycling track may be reduced to allow retrofitting of the track to existing site conditions for a certain distance. A transition track taper is selected to ensures a gradual change of the track width.

3. CROSSING OF EXISTING ROADS

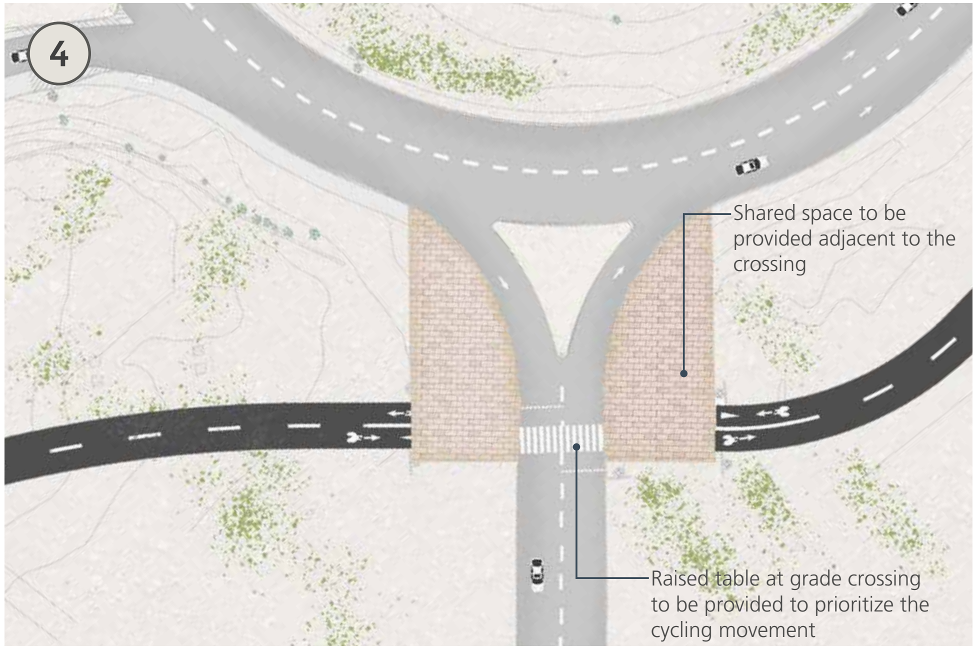
In line with current Abu Dhabi standards, shared space may be provided adjacent to crossings of the existing road. A raised at grade crossing may be provided to prioritize the cycling movement, as well as providing traffic calming for motor vehicles.

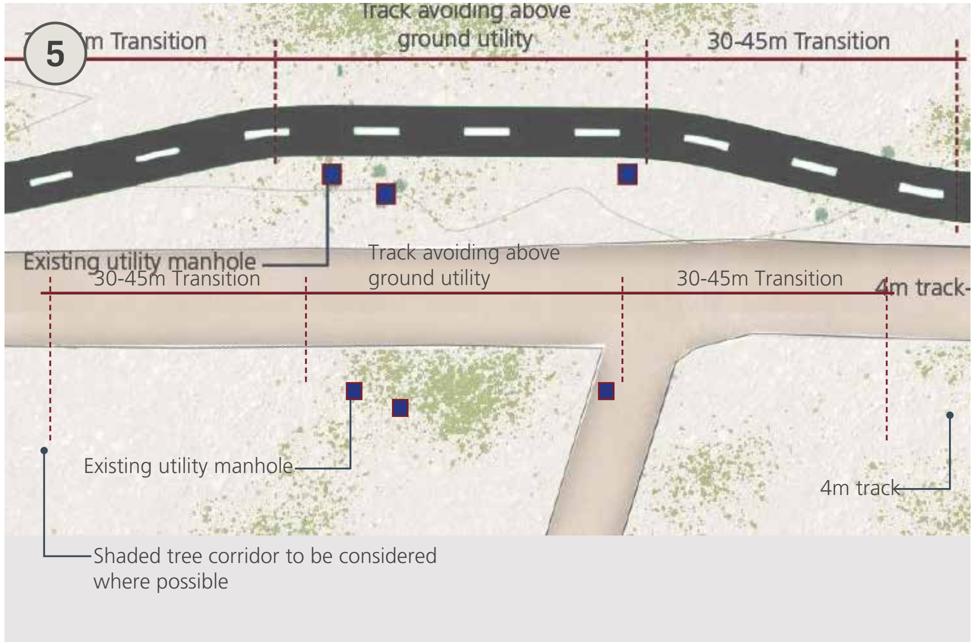
4. CROSSING OF EXISTING ROUNDABOUTS OR JUNCTIONS

Like the crossing with existing roads, the cycle track crossing existing roundabouts will also require a provision of a shared space, either for existing or additional if there are no nearby existing facility.

5. AVOIDING UTILITIES

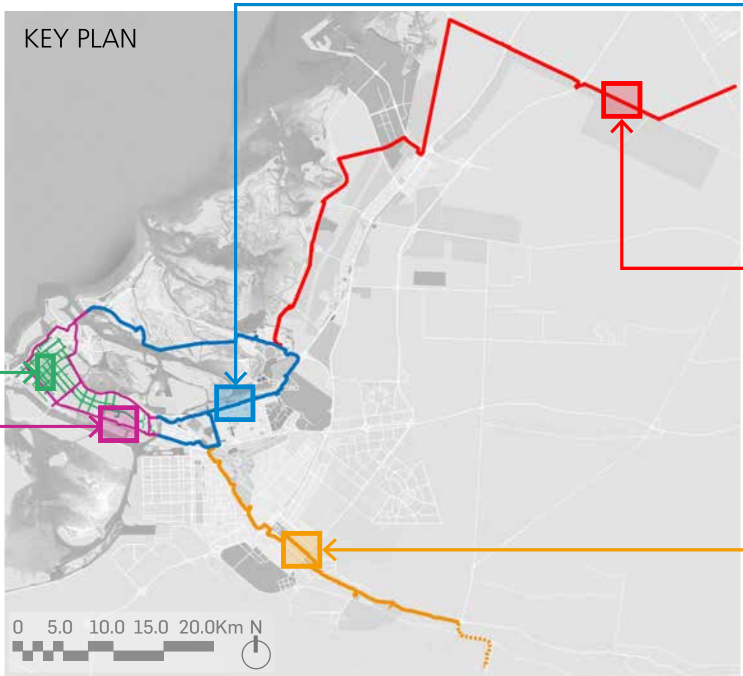
Since the cycling track must be within the available right of way or within public space outside of any plot boundary, there are locations where the existing utility conflicts with the alignment. To avoid any clash in these instances, an adjustment to the track may be proposed.





ABU DHABI ISLAND AT-GRADE CYCLE NETWORK

ABU DHABI ISLAND HIGH-SPEED CYCLE TRACK



ABU DHABI MAINLAND (CONNECTION OF AD ISLAND TO YAS ISLAND AND TO SAADIYAT ISLAND)

ABU DHABI MAINLAND TO ABU DHABI BOUNDARY (AL QUDRA)

ABU DHABI MAINLAND TO SOUTH OF BANIYAS (AL WATHBA)

3 THE DESIGN PROCESS

3 THE DESIGN PROCESS

KEY PROCESS FLOW AND STAGES

The process by which the ADCN and its constituent routes will be planned, designed, constructed, and commissioned is shown in Figure 17, including indicating the Authority Approvals and No Objections required at each stage.

The process flow consists of seven stages: Pre-Concept Design, Concept Design, Preliminary Design, Construction, Commissioning, and Operation and Management.

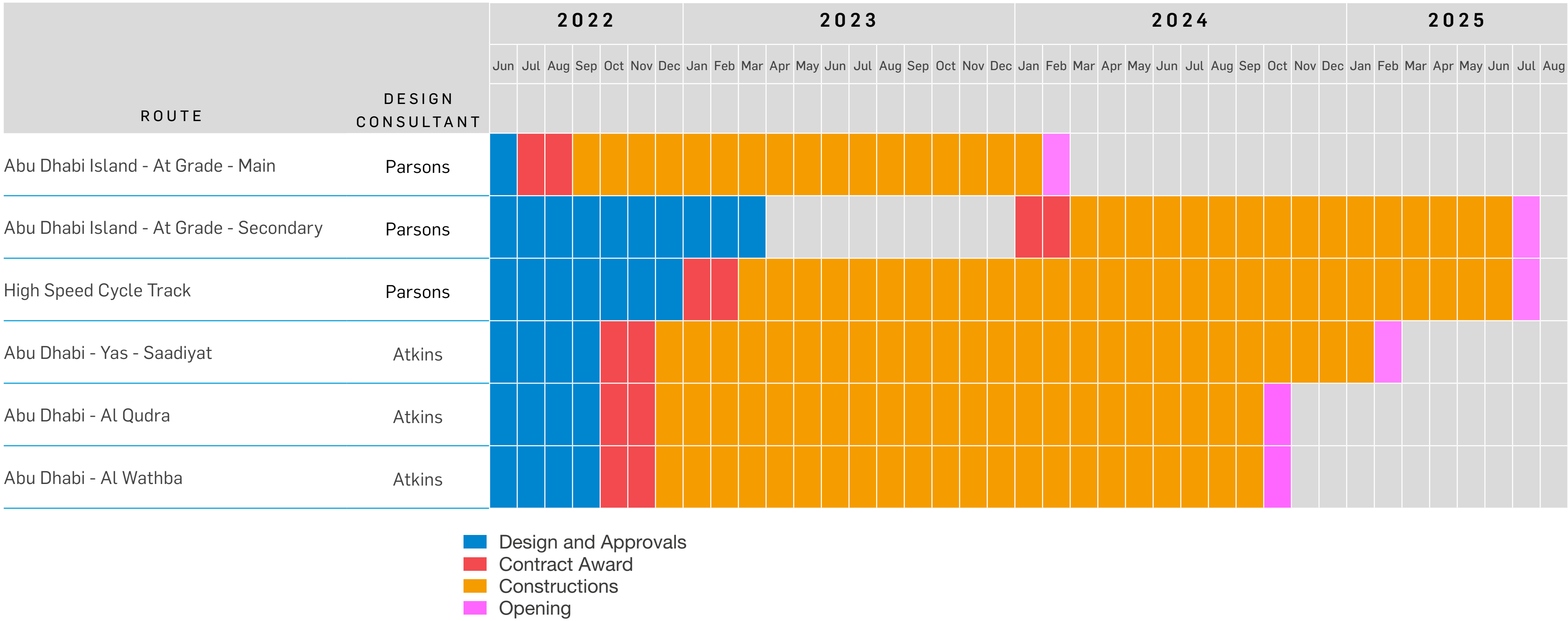
In each stage key considerations, products, and an indicative list of necessary stakeholder approvals and NOCs are considered, variable by ADCN route and section.

INDICATIVE TIMESCALES AND PROGRAM

Applying the key stages, delivery of the full ADCN across all five routes is intended be complete by mid-2025. The key project stages by route are shown in Table 6.

Individual phases within routes may be opened to the public within these timelines.

Table 6: Project Program and Milestones



	00 Pre-Concept Design Stage	01 Concept Design Stage	02 Preliminary Design Stage	03 Detailed Design Stage	04 Construction Stage	05 Commissioning Stage	06 Operation & Maintenance Stage
Considerations	• Definition of vision and objectives • Definition of users and their requirements • Site context analysis • Benchmarking • Optioneering of alignment • Route and node definition • Route alignment • Route themes/typologies and cross-sections • Data validation • Node locations and initial concepts • Node typologies	• Route layout plans to concept design level of detail • Public realm landscape definition • Utility requirements: modification, protection and relocation strategy • Wayfinding • Node design • Landscape design • Architecture and buildings design • Parking area requirements • Waste management • Onwani requirements • Clashes with other projects • Project cost estimation • Traffic impact plan • Stage 0 Road Safety Audit	• Refinement of route layout plans to preliminary design level of detail • Calculations of technical details for bridges and other constructions • Marine engineering • Traffic count • Land acquisition needs • Verification of tree locations and relocation strategy • Waste management • Emergency accessibility strategy and assessments • Geotechnical surveys • Architectural details • Fire life safety strategy • Traffic impact assessment • Stage 1 Road Safety Audit	• Refinement of route layout plans to detailed design level: roads, cycling track, intersections, infrastructure, utilities, landscape, architectural elements, street furniture • Detailed topographical, geological and geotechnical surveys • Land acquisition • Supporting cycling strategy and action plan • Maintenance access plan • Branding of the routes • Traffic impact assessment • Stage 2 Road Safety Audit	• Detailed construction time plan and workflow charts for individual sections • Site work preparation • Utilities protection • Utilities demolition/ diversion • Construction of infrastructure (bridges, ramps, junctions etc.) • Construction of cycle track • Construction of landscape • Construction of nodes • Supervision of construction works • Construction work and infrastructure quality warranties	• Project commissioning plan • Approvals and quality assurance tests • Asset management plan • Cycling event schedule • Communication plan • Public launch plan • Opening phasing plan • Stage 3 Road Safety Audit	• Maintenance and asset management procedures • Preventive and active maintenance • Maintenance inspections • Operation and maintenance services • Monitoring of user impact • Improvement assessment plans • Stage 4 Road Safety Audit
Products	Pre-concept report drawings, sketches, diagrams	Concept design report and drawings, e.g., route horizontal alignment, and cross-sections, public realm strategies	Preliminary design report and detailed drawings accomodating utilities, landscape, alignment, road and water crossing details	Detailed design report and detailed drawings	Completion of bicycle tracks and supporting infrastructures	Approvals, plans, commissioning report	Plans, schedules, and inspections
Stakeholder Approvals and NOCs	• DMT: Planning approval • Smart Hub - ADM: Roads Geometry, Landscape, Utilities Protection, Storm water, Irrigation, Lighting • Service providers: Power & Water, Sewerage, Telecom Network	• ITC: Traffic impact plan • Smart Hub - ADM: Roads Geometry, Landscape, Utilities Protection, Storm water, Irrigation, Lighting • Service providers: Power & Water, Sewerage, Telecom Network	• ITC: Traffic impact assessment • Smart Hub - ADM: Roads, Irrigation protection, Lighting, Stormwater protection, Landscape. • Service providers: Telecommunications, Power & Water, Sewerage, Transco, ADNOC.	• ITC: Traffic impact assessment • Smart hub - ADM: Roads, Utilities protection, Irrigation protection, Lighting, Stormwater protection, Landscape. • Service providers: Telecommunications, Power & Water, Sewerage, Transco, ADNOC.	• DMT • Smart Hub - ADM: Roads, Utilities protection, Irrigation protection, Lighting, Stormwater protection, Landscape. • Service providers: Telecommunications, Power & Water, Sewerage, Transco, ADNOC.	• DMT • ADM • Service providers	• DMT • ADM • Service providers

Figure 17: Key ADCN Design Process Flow and Stages

4 MONITORING THE BENEFITS

4 MONITORING THE BENEFITS

MONITORING DELIVERY AND OUTCOMES

The vision defined in Chapter 1 has desired outcomes around cycle usage, recreation, safety and integration aligned with the DMT priorities of Active Environment, Active Knowledge, Active System and Active People.

The outcomes can be measured through Key Performance Indicators (KPIs) which are Specific, Measurable, Attainable, Relevant, and Time-Bound (SMART). Proposed KPIs are shown in Table 7, starting from and extending the measures already documented in DMT's Active Recreation and Cycling Strategy.

Table 7: Key Performance Indicators for Monitoring Impacts of the Abu Dhabi Cycling Network

PRINCIPLE	KEY PERFORMANCE INDICATORS
1. USAGE AND INFRASTRUCTURE	**Primary Indicator** Change in volume of cyclists passing selected locations or cordons, or using specific routes **Secondary Indicators** % mode share for cycling by group, trip purpose and length Length, scale and classification of new cycling infrastructure and facilities, overall/per population % Infrastructure assets well managed and maintained
2. SAFETY	**Primary Indicator** Number of cyclists killed or seriously injured in road accidents overall and per 100,000 cyclists **Secondary Indicators** User perceptions of safety and personal security when cycling (Survey) Number of cycles stolen or subject to vandalism overall and per 100,000 cyclists Number and classification of penalties and fines issued for violations of applicable regulations
3. EVENTS, ACTIVE AND HEALTHY LIFESTYLES	**Primary Indicator** Number of registered participants in organised cycling events, training, and awareness activities **Secondary Indicators** Public knowledge and perception of Abu Dhabi cycling initiatives, investment, and promotion % of children, by age group, cycling as part of daily activities Changes in specified public health outcomes, such as obesity, cardiac conditions, and diabetes
4. INTEGRATION AND SHARING	**Primary Indicator** Number of shared bicycles and sharing stations, overall and per population **Secondary Indicators** Number and share of public transport passengers using bicycles as a feeder mode Number of eligible bicycles carried on public transport vehicles
5. GOVERNANCE	**Primary Indicator** Number of collaborative mechanisms to promote cycling between public and private sectors **Secondary Indicators** Adherence to infrastructure and other delivery plans Capacity/skills of private sector supply chain for cycling infrastructure, equipment, and services

Once the KPIs are accepted as the basis for monitoring outcome delivery, the following steps are required:

- Develop KPI definitions and data collection methodologies, with supporting budgets and resources;
- Establish baseline levels of performance in the current situation;
- Define targets and desired directions of travel for KPI movement;
- Develop a focused and suitable data analysis and reporting platform; and
- Establish governance arrangements for reviewing, assessing and performance managing delivery as a result of KPI data and analysis of progress.

Efforts should also be undertaken to develop smart sensors and platforms so that data collection and population of KPIs can be done automatically, building a database of performance, saving time and cost.

Separate KPIs for the detailed design and delivery of the ADCN routes by the selected Design-Build Contractors in terms of program, budget and quality are being developed to ensure efficient and timely implementation in line with the Abu Dhabi Cycling vision.

APPENDICES

APPENDIX A – TERMINOLOGY

KEY ACRONYMS

AD	Abu Dhabi
ADCC	Abu Dhabi Cycling Club
ADCD	Abu Dhabi Civil Defense
ADCN	Abu Dhabi Cycling Network
ADDC	Abu Dhabi Distribution Company
ADEK	Department of Education and Knowledge
ADEO	Abu Dhabi Executive Office
ADI	Abu Dhabi Island
ADM	Abu Dhabi Municipality
ADNEC	Abu Dhabi National Exhibition Centre
ADNOC	Abu Dhabi National Oil Company
ADQCC	Abu Dhabi Quality and Conformity Council
ADSC	Abu Dhabi Sports Council
D&B	Design and Build
DCD	Department of Community Development
DCT	Department of Culture and Tourism
DMT	Department of Municipalities and Transport
DoH	Department of Health
EAD	Environment Agency Abu Dhabi
EVA	Emergency Vehicle Access
GHQ	UAE Police General Headquarters
HSCT	High Speed Cycle Track
ITC	Integrated Transport Centre
KPI	Key Performance Indicator
SMART	Specific, Measurable, Attainable, Relevant, and Time-Bound
TRANSCO	Abu Dhabi Transmission and Despatch Company
UCI	Union Cycliste Internationale

KEY DEFINITIONS

BICYCLE	A small human-powered or motor-powered vehicle with two or more wheels for personal transportation of people and goods, propelled on a public right-of-way by means other than fossil fuels.
CYCLE TRACK	A dedicated off-road facility designed and reserved for cycling that can be either one-way or two-way.
CYCLE LANE	A one-way cycle facility marked on a road surface which travels with flow of traffic within the Right of Way which demarcates space on the road reserved for cycling.
SHARED PATH	An off-road facility which separates pedestrian and cycle movement with use of a means of demarcation, such as painted line, to define the user space.
USER GROUPS	Defined groups of users, with differing needs and requirements, who will ride on the designated cycle infrastructure.
ROUTE TYPOLOGY	Route typology is the classification of sections of the route by the combination of user groups most likely to be utilising each respective section and the associated facilities.

APPENDIX B – APPLICABLE STANDARDS

NO.	DISCIPLINE	STANDARD	VERSION	DATE
1	International	UCI Cycling Regulations. Part 2 Road Races and Organiser's Guide to Road Events		2022
2		Design Handbook for Bicycle Traffic, Denmark		2014
3		Best Practices Dutch Cycling, Dutch Cycling Embassy, The Netherlands		2017
4		Geometric Design Parameters for Cycling Infrastructure, European Cyclists' Federation		2022
5		London Cycling Design Standards, UK		2019
6		Cycle infrastructure design (Local Transport Note 1/20), UK		2020
7		Design Manual for Roads and Bridges (DMRB), UK		
8		Street Design Manual: Walkable Neighborhoods, Queensland, Australia		2020
9	General	Abu Dhabi Quality and Conformity Council (ADQCC). Abu Dhabi Emirate Guideline for Infrastructure Services Standards	Second Edition	2017
10		Integrated Transport Centre – E-Scooter - Bike Signage		
11	Landscape Architecture	Abu Dhabi Urban Street Design Manual (USDM) (ROW-603)	1.1	
12		Public Realm Design Manual (PRDM) (PR-401)	-	2018
13		Abu Dhabi Walking and Cycling Master Plan (WCMP) - Network Design (TR-530)	-	2011
14		Abu Dhabi Cycling Network 2025	-	2018
15	Roads	Road Geometric Design Manual (DMAT) (TR-514)	Second Edition	2021
16		Manual on Uniform Traffic Control Devices (TR-511)	Second Edition	2020
17		Roadside Design Guide (TR-518)	First Edition	2016
18		Pavement Design Manual (TR-513)	Second Edition	2021
19		Standard drawings (TR-541 - 1 & 2)	ADQCC website – Second Edition	2022
20		Technical Circular (UPC/PI/BL/EDO/401) (Updates to USDM)	-	2017
21		Technical Circular issued by Executive Council (17/020/2/2018)	-	2018
22	Roads	Abu Dhabi Paving Strategy	-	2021
23	Cycle Path Lighting	Abu Dhabi Lighting Manual PR-402	Third Edition	2021
24	Structures	ADQCC Road Structures Design Manual TR-516	Second Edition	2021
25		ADQCC Standard Construction Specifications TR-542 Part 1 and Part 2	First Edition	2018
26		AASHTO LRFD Bridge Design Specifications	9th Edition	2020
27		AASHTO LRFD Guide Specifications for the Design of Pedestrian Bridges	Second Edition	2009
28		AASHTO Standard Specifications for Structural Supports for Highway Signs, Luminaires, and Traffic Signals	Sixth Edition	2013
29		AASHTO Guide for the Development of Bicycle Facilities	Fourth Edition	2012
30		CG300 Technical Approval of Highway Structures		
31		PTI DC45.1-12 Recommendations for Stay Cable Design		
32		SETRA, Technical Guide for Footbridges		
33		Brief Dutch Design Manual for Bicycle and Pedestrian Bridges		2017

APPENDIX C – LIST OF REFERENCES

[1] Road Geometric Design Manual (DMAT) (TR-514). Note-2 (p. 688)

[2] Road Geometric Design Manual (DMAT) (TR-514). Clause 2.7.4.8 (p. 123)

[3] AASHTO Guide for the Development of Bicycle Facilities. Table 5.2 (p. 5-14)

[4] AASHTO Guide for the Development of Bicycle Facilities (p. 5-6)

[5] Abu Dhabi Tiles Strategy 2018-2030

[6] Cycle Infrastructure Design, DOT UK. Clause 5.9.7 (p. 46) (Recommendation based on optimization, considering best practice experience.)

[7] AASHTO Guide for the Development of Bicycle Facilities (p. 5-5) (Recommendation based on optimization, considering best practice experience.)

[8] Abu Dhabi Walking and Cycling Master Plan (WCMP) - Network Design (TR-530). Table 6.10. (Recommendation based on considering best practice experience.)

[9] Abu Dhabi Lighting Manual PR-402, Table no. 4.3.1 (p. 19)

[10] Recommendation based on considering best practice experience.

[11] Manual on Uniform Traffic Control Devices (TR-511). Chapter 8

[12] Traffic Signs and Marking of Abu Dhabi Walking and Cycling Master Plan (WCMP) - Network Design (TR-530). Chapter 2, Section 2

[13] Cycle Infrastructure Design. DOT UK. Chapter 10

[14] Abu Dhabi Walking and Cycling Master Plan (WCMP) - Network Design (TR-530). Section 7.3

[15] Standard Drawing Guideline part 1 TR 541-1. Drawing GM 23 (p. 48)

[16] Abu Dhabi Walking and Cycling Master Plan (WCMP) - Network Design (TR-530). Section 3.2

[17] Standard Drawing Guideline part 1 TR 541-1. Drawing GM 19

[18] Cycle Infrastructure Design, DOT UK. Table 5-5 (p. 44)

[19] Cycle Infrastructure Design, DOT UK. Table 5-7 (p. 46)

[20] Road Geometric Design Manual (DMAT) (TR-514). Section 7.4.9 (p. 342)

[21] For Navigational clearance, Information received from Abu Dhabi Maritime.

[22] Brief Dutch Design Manual for Bicycle and Pedestrian Bridges. Section 3.1 (p. 26)

[23] Abu Dhabi Walking and Cycling Master Plan (WCMP) - Network Design (TR-530) (p. 12)

[24] Brief Dutch Design Manual for Bicycle and Pedestrian Bridges. Section 3.3 (p. 28)

[25] Recommendation based on optimization, considering best practice experience & visual barriers for the building occupant

[26] Brief Dutch Design Manual for Bicycle and Pedestrian Bridges. Section 2.4 (p. 22) (Suggests 2.5m. 2m recommended based on optimization considering best practice experience)

[27] Brief Dutch Design Manual for Bicycle and Pedestrian Bridges Section 5.1 (p. 50) (Suggests 1.2-1.3m. 1.5m adopted based on best practice experience)

[28] Road Geometric Design Manual (DMAT) (TR-514). Table 13-2

[29] Road Geometric Design Manual (DMAT) (TR-514). Section 13.3.4.5

[30] Road Geometric Design Manual (DMAT) (TR-514). Section 13.3.4.6

[31] Road Geometric Design Manual (DMAT) (TR-514). Section 6.5

[32] Standard Drawing Guideline part 1 TR 541-1. Drawing GM 22

[33] Abu Dhabi Walking and Cycling Master Plan (WCMP) - Network Design (TR-530). Section 3.15

[34] Road Geometric Design Manual (DMAT) (TR-514). Section 13.3.4.3

[35] Abu Dhabi Lighting Manual PR-402. Part 2- Section A (p. 457)

ALDAR

مكتب أبوظبي التنفيذي
ABU DHABI EXECUTIVE OFFICE

دائرة البلديات والنقل
DEPARTMENT OF MUNICIPALITIES AND TRANSPORT

دائرة تنمية المجتمع
DEPARTMENT OF COMMUNITY DEVELOPMENT

Abu Dhabi Sports Council
مجلس أبوظبي الرياضي

ABU DHABI CYCLING CLUB نادي أبوظبي للدراجات

PARSONS

ATKINS
Member of the SNC-Lavalin Group